



## Effect of Nano Amino Acids and Soaking Period on Germination Characteristics and Nutrient Allocation in *Panicum Maximum* L.

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**Abstract:** This study aimed to investigate the impact of nano amino acids on seeds germination and early seedling development of *Panicum maximum* L. The experiment was carried out at the laboratories of the Center of Desert Studies, University of Anbar based on a factorial randomized complete block design (RCBD) with four replications. The experiments included two periods of soaking (5 and 10 hours) and three concentrations of nano amino acids solutions (0, 0.75, and 1.5 mL L<sup>-1</sup>). The results showed that the N1 concentration (0.75 mL L<sup>-1</sup>) produced the best germination, primary radical plumule lengths, and seedling dry weights compared to the control treatment and higher concentrations. The study demonstrates a consistent and stabilized distribution of nitrogen, carbon, and oxygen as determined by SEM–EDS elemental mapping in the seedling tissues for the N1 treatment. This homeostasis was accompanied by effective nutrient absorption, enhanced metabolism and improved physiological functions. Conversely, the osmotic and ionic stress induced by the higher N2 concentration (1.5 mL L<sup>-1</sup>) served to disturb water potential gradient causing reduced germination and vigor. The priming period is highly important in terms of the observed vigor indices as the 10 h application (T2) significantly increased germination through improved hydration, enzyme activation, and nutrient uptake. Moderate concentrations of nano amino acids combined with prolonged soaking hours promoted seed germination, supported strong seedling vigor and ensured a balanced nutrient distribution. The results of this study highlight nano priming as a promising and environment-friendly approach for improving seedling and early plant establishment performances in tropical forage species including *Panicum maximum*. It presents evidence that differs from prior nano-priming examples by utilizing nano amino acids and associating physiological seedling responses with an SEM–EDS elemental analysis using *Panicum maximum*, a forage grass that has been less studied with respect to nano amino acids application.

**Keywords:** nano amino acids, seed priming, *Panicum maximum* L., germination percentage, SEM–EDS analysis

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### Introduction

Germination is of vital importance in the first growth and crop production of a plant and represents a crucial stage in its life cycle. Differences in germination and early tolerance determine seedling establishment success, highlighting the importance of proper seed activation (13,20,22). Nano amino acids are of huge importance in sustainable agriculture due to their excellent ability in enhancing nutrient uptake and promoting physiological processes in plants. Germination in a promising environment in the cultivation of agricultural seeds is primed in nano amino acids before planting, and it is a promising growth environment where the particles help sprout and develop higher quality seeds (26).

The effect of physiological fertilizers on seed germination and field establishment of early growth of seedlings is an influential part of sustainable agriculture, especially in regions that suffering constant environmental pollutants (5, 17, 25). Soil often serves as a reservoir for dangerous pollutants with heavy metals within it acting as highly toxic contaminants causing serious damage to seed germination and seedling development, and compromising global food supply and security (11, 31). Nanoparticle priming (nano-priming) is a seed treatment that increases the metabolic and biochemical activities of seeds. It induces small holes in the coat that allow the plant to take in more oxygen (5, 23). It also helps with the antioxidant defense mechanisms for seeds (ROS) and

generates hydroxyl radicals to loosen cell walls which further contributes to the breakdown of starch (40, 41), as well as utilizes the physiological and biochemical changes to its advantage for germination, stronger seedlings, and superior yields.

Nano-seed priming as demonstrated has been shown to enhance plant growth and productivity (16, 24). The method has various advantages such as higher germination rates, better seedling vigor, and increased crop yield (20). Effects of nanoparticles inside plant tissues play a vital role in promoting biological functions including photosynthesis through its effect on the vegetative part during the early stages of germination and nitrogen metabolism (3, 33). Similarly, evaluations of nano amino acids show their alignment with national and global agricultural sustainability aspirations that provide a roadmap for farming. This is a pathway for them to go in genetics modification and plant breeding. Nevertheless, despite their numerous benefits in terms of food safety and quality, there are risks associated not only with human health but also animals as well as environmental beings (10, 14, 29).

This study aimed to measure germination performance, seedling vigor, and essential allocation (C, N, and O) under different nano amino acids concentrations and priming periods using SEM–EDS analysis.

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## Materials and Methods

This study was conducted at the laboratories of the Centre of Desert Studies, University of Anbar (2025) as a factorial experiment based on the randomized complete block design (RCBD) with four replicates. The results were analyzed using the GenStat program.

Factor A (nano amino acids) – NP: containing 30% organic matter [amino acids], 3% organic nitrogen, and 5% mineral nitrogen. The acids were manufactured using Turkish nanotechnology at three levels:

- N0 = 0 mL L<sup>-1</sup> (control)
- N1 = 0.75 mL L<sup>-1</sup>
- N2 = 1.5 mL L<sup>-1</sup>

Factor B, involving two soaking periods, namely:

- T1 = 5 hours
- T2 = 10 hours

The treatments were randomly allocated within a block to reduce the variability of experimental units. The purpose was to determine the effects of concentration and soaking period on seed germination and establishment and seedling performance of *Panicum maximum* L.

### Preparing and germinating seeds

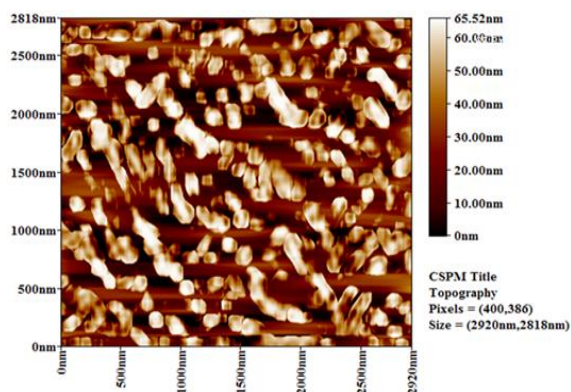
Seeds were washed three times with distilled water after 10 min sterilization in 10% sodium hypochlorite. Nano amino acids suspensions were dipped at the appropriate concentrations while those in the control group were dipped in distilled water and soaked at room temperature (20 °C) for 24 h under dark conditions. The primed seeds were washed with distilled water for 2 min, surface-dried with filter paper, and air-dried at 25–30 °C to about 10% moisture (20). Two hundred seeds from each treatment were sown in Petri dishes (9 cm diameter) with four replications of 50 seeds each. The germination test was carried out at 25 ± 2 °C and the number of germinated seeds was noted daily for ten days (33).

### Measured Traits

1. Germination percentage: Standard laboratory germination test (final count) - natural germination was calculated only after the ten-day incubation period (15), and the results were converted to percentages using the formula:

$$\text{Germination \%} = (\text{Number of natural germination after 10 days} / \text{Total number of seeds sown}) \times 100$$

2. Radicle and plumule length (cm) in standard germination test: Ten normal seedlings were randomly removed from



**Figure 1.** Nanoscale topographic characterization of nano amino acids by AFM.

the test at the end of the test period (ten days), followed by removal of both plumule and radicles, from its point of running from their attachment to the seed. The length of radicle and plumule were measured in centimeters with a ruler (15).

3. Seedling dry weight (g): A random sample of ten normal seedlings were selected after the test duration (ten days) by separating radicle and plumule from their point of attachment up to the seeds. Both were then packed in a perforated paper bag and dried at 80 °C for 24 h. The mean seedling dry weight was determined with a 4-decimal place electronic balance and was derived by dividing the total weight of dry seedlings by their number (15).
4. Seedling vigor = (Germination %) × (Radicle Length + Plumule Length) (32).
5. Vigor index of seedlings = Germination Rate × Seedling Dry Weight (32).

### SEM-EDS Analysis

The elemental composition of the *Panicum maximum* L. seedlings was analyzed using scanning electron microscopy equipped with energy-dispersive X-ray spectroscopy (SEM-EDS) at the Industrial Research and Development Authority (Iraq). Fresh root and shoot tissues were collected 10 days after germination and cut into small pieces (2–5 mm). The samples were gently rinsed with distilled water to remove surface contaminants and blotted dry using filter paper.

Fresh samples were used to avoid chemical alterations associated with fixation and dehydration, ensuring accurate elemental detection. The samples were mounted on aluminum stubs using carbon tape and coated with a thin carbon layer to improve conductivity and minimize charging during imaging. SEM analysis was performed at an accelerating voltage of 15 kV and a working distance of approximately 10 mm. Elemental analysis (C, N, and O) was conducted using EDS with ZAF correction based on multiple measurement points per sample. Measurements were performed on five independent biological replicates (n=5) to ensure accuracy and reproducibility (38).

### AFM Analysis

Fresh *Panicum maximum* L. seedling tissues (root and shoot) were collected 10 days after germination and processed immediately without chemical fixation or dehydration to preserve their native physicochemical characteristics. The samples were gently rinsed with distilled water and blotted dry, then sectioned into small pieces (2–5 mm). For AFM analysis, separate fresh samples were prepared and analyzed under ambient conditions. Measurements were conducted on five independent replicates (n = 5) to ensure data reliability.

The obtained AFM images were processed using Gwyddion software to obtain quantitative surface parameters, including roughness and particle size distribution, based on grain analysis and height profile measurements (Figure 1) (27).

## Results and Discussion

### Germination Percentage

Table 1 shows the effect of soaking period and nano amino acids concentrations on germination percentage. The highest mean germination percentage (85.38%) was in the N1 treatment (0.75 mL L<sup>-1</sup>), which was significantly different

from N0 and N2 based on LSD test ( $P \leq 0.05$ ), whereas the interaction of the soaking period  $\times$  nano amino acids concentrations had no significant impact. This enhancement under N1 can be attributed to the well-proportioned nutrient supply. The SEM-EDS analysis indicated a homogeneous distribution of C, N, and O in the *Panicum maximum* tissues, and the higher concentration of nano amino acids reduced the influx of carbon (C) content (Table 8), while N1 favored a greater uptake of nitrogen (N) compared with the other nano amino acids concentrations (Table 7). This is because nitrogen and carbon are closely linked in plant metabolism — as more nitrogen accumulates, more of it goes toward synthesizing proteins and amino acids, with less carbon left behind to be stored as carbohydrates.

The reduction in carbon does not reflect carbon having been lost from the plant, rather that it has actively been incorporated into metabolism, for use as energy and structural material to develop a healthier, faster-growing seedling (Figure 2). The match up distribution of C, N, and O elements under the N1 treatment is perhaps connected with increased metabolic activity and growth-related activities. Though phytohormones were not directly assessed in this study, earlier reports indicate that enhanced nutrient balance and nano-priming cannot directly increase hormonal regulation related to early seedling growth (e.g., gibberellins, auxins, and cytokinins) (8). Furthermore, the small size of the nano-particles makes it easier for them to pass through seed coats and improve the gaseous exchange in seeds (Table 9). This surplus oxygen stimulates respiration and metabolic processes, breaking seed dormancy and accelerating germination. These findings are consistent with those of (19) in which all stated that nano amino acids led to significant enhancement seed germination and early vigor.

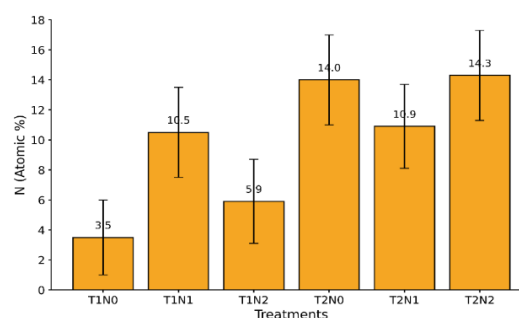
Earlier works on this subject (5) noted that nano amino acids with high nitrogen content had better seed germination as well as initial seedling vigor via improved nutrient availability and uptake efficiency (6). These results match current findings showing higher N1 performance in terms of germination and adequate allocation of nitrogen to tissue in *Panicum maximum*. Accordingly, high concentrations of nano amino acids, particularly when rich in nitrogen and transformed to ammonium and nitrate, have the potential to be toxic for seed tissues. This toxicity may be linked with disturbances in ionic homeostasis and cellular pH, as noted in earlier studies, which can negatively influence germination processes. The primary mechanism for this effect is osmotic stress, since an elevated level of nanoparticles in the soaking solution results in higher osmotic pressure at the seeds (Figure 3). Thus, excessive nano amino acids levels inhibited germination, mainly due to osmotic, ionic, and oxidative stresses, as reported previously (1, 28), who also emphasized the deleterious effects of high nanoparticle levels on plant growth and seed health. These authors also demonstrate adverse effects of nano-particles at a high concentration.

With regard to the priming period, treatment T2 (18 h) resulted in the greatest rate of germination (79.67%), presenting also 8.67% more than T1. This enhancement is due to the longer soaking duration, which provided enough time for the seeds to up take water and achieve total imbibition. This activated internal enzymes and helped degrade certain inhibitors that otherwise would have delayed or precluded germination. In addition, the long soaking affected better seed

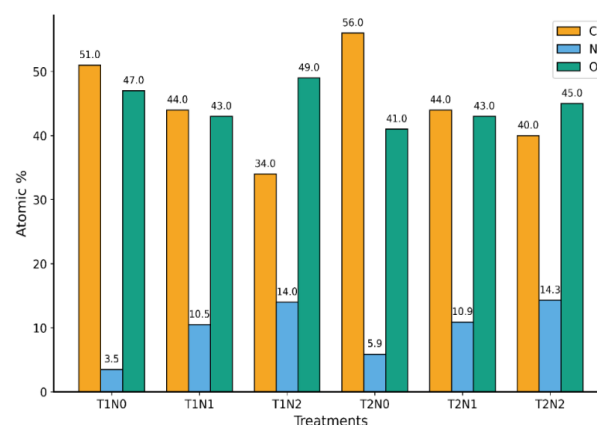
**Table 1.** Effect of nano amino acids seed priming on germination percentage.

| Period Treatment | Nano Treatment (mL L <sup>-1</sup> ) |              |              | Mean  |
|------------------|--------------------------------------|--------------|--------------|-------|
|                  | N0 (0)                               | N1 (0.75)    | N2 (1.50)    |       |
| T1               | 67.25                                | 79.25        | 66.50        | 71.00 |
| T2               | 78.75                                | 91.50        | 68.75        | 79.67 |
| Mean             | 73.00                                | 85.38        | 67.63        |       |
| LSD              | Period * Nano<br>5.56                | Nano<br>6.81 | Period<br>NS |       |

Values are means of four replicates. LSD values Were calculated at  $P \leq 0.05$ .



**Figure 2.** Effect of seed activation with nano amino acids on nitrogen content in *Panicum* tissue after 10 days of planting (SEM-EDS)

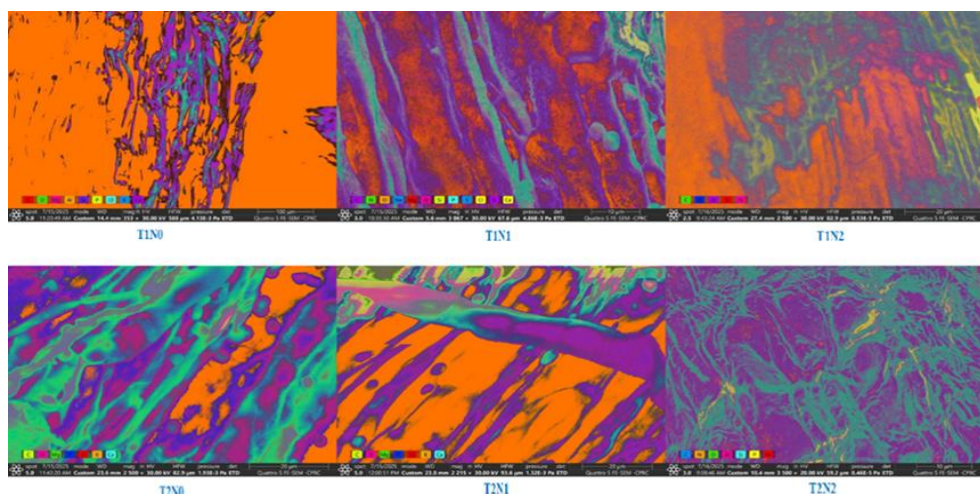


**Figure 3.** Effect of seed activation with nano amino acids on carbon, nitrogen, and oxygen content in seedling tissue after 10 days of planting (SEM-EDS)

usage of the available nutrients in seeds yielding a high accumulation of dry matter (Table 4) which directly serves as a facilitator for both radicle and plumule growth (Tables 2 and 3). However, no significant interactions were found between the factors under study for this trait.

### Radicle Length (cm)

The statistical analysis showed significant differences with nano amino acids activation treatments and soaking durations and a nonsignificant interaction between both factors (Table 2). The average radicle length of 11.188 cm for N1 with nano amino acids treatment was significantly higher over N2 and N0, with deviations of 3.188 cm and 2.563 cm respectively, though the two latter treatments did not differ significantly from each other. These results match the findings of (21) that nano-material seed treatments had a positive effect on radicle and seedling growth. Those findings showed that seed coating with nano-particles was beneficial for radicle growth of the plants since nutrient availability was increased (Figure 4). The



**Figure 4.** SEM–EDS elemental mapping of *Panicum maximum* L seedling tissues under seed priming and nano amino acids treatments.

superior efficacy of N1 could be due to the effect of nitrogen (Table 7) and nano-form amino acids directly interacting with them, which collectively triggers the activity of growth hormones such as gibberellins, auxins, and cytokinins. Furthermore, the excellent penetrability of nanoparticles allowed them easier entry into the seed coat and to accumulate more oxygen (Table 9) within the seed, promoting metabolic activity.

This activity is an initiator for the production of enzyme proteins that degrade reserve carbohydrates and lipids, but also of structural proteins necessary to build new cells in the root meristem. SEM–EDS analysis suggested a relatively balanced composition of carbon/nitrogen/oxygen elements in the N1 treatment. The observed balance of the three elements may indicate improved nutrient availability and metabolic activity, although SEM–EDS provides indirect evidence. Increased nitrogen availability may enhance hydrolytic enzyme activity (e.g., amylases and lipases), which could be confirmed by enzyme assays, gene expression, or metabolite analysis. The concomitant decrease of the radicle length at higher nano amino acids concentrations may be explained by the toxic effect in seed tissues due to high amounts of nano-elements especially nitrogen (Table 7), as they can turn into ammonium and nitrates.

Extremely high concentrations of nano amino acids might disrupt ionic balance and change the cellular pH, as suggested in earlier studies, therefore affecting metabolic processes inside the plant cell. This study did not directly measure these effects but only inferred them based on the literature that already established these effects. These results agree with (12) which emphasized the adverse influence of excessive nano-material doses on plant tissues.

With respect to soaking period, the average radicle length of seedlings for treatment T2 was the highest at 10.292 cm, which was significantly different at 2.042 cm from that of T1. The superior results of T2 may be due to the longer soaking duration enabling the seeds to reach the imbibition stage of germination with sufficient period for uptake. This initiated the dormant enzymes and consequently such enzymatic activation increased metabolic activity and utilization efficiency of reserves. Furthermore, the longer soaking enriched the water with primary nutrients as evidenced by dry seedling weight (Figure 7), and hence produced stronger radicle length growth.

**Table 2.** Effect of nano amino acids seed priming on radicle length (cm).

| Period Treatment | Nano Treatment (mL L <sup>-1</sup> ) |               |              | Mean   |
|------------------|--------------------------------------|---------------|--------------|--------|
|                  | N0 (0)                               | N1 (0.75)     | N2 (1.50)    |        |
| T1               | 7.000                                | 9.750         | 8.000        | 8.250  |
| T2               | 9.000                                | 12.625        | 9.250        | 10.292 |
| Mean             | 8.000                                | 11.188        | 8.625        |        |
| LSD              | Period * Nano<br>1.749               | Nano<br>2.142 | Period<br>NS |        |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

**Table 3.** Effect of nano amino acids seed priming on plumule length (cm)

| Period Treatment | Nano Treatment (mL L <sup>-1</sup> ) |              |              | Mean  |
|------------------|--------------------------------------|--------------|--------------|-------|
|                  | N0 (0)                               | N1 (0.75)    | N2 (1.50)    |       |
| T1               | 6.875                                | 10.625       | 8.250        | 8.583 |
| T2               | 9.125                                | 12.625       | 7.625        | 9.792 |
| Mean             | 8.000                                | 11.625       | 7.938        |       |
| LSD              | Period * Nano<br>1.069               | Nano<br>1.31 | Period<br>NS |       |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

**Table 4.** Effect of nano amino acids seed priming on dry seedling weight (g)

| Period Treatment | Nano Treatment (mL L <sup>-1</sup> ) |               |              | Mean  |
|------------------|--------------------------------------|---------------|--------------|-------|
|                  | N0 (0)                               | N1 (0.75)     | N2 (1.50)    |       |
| T1               | 0.180                                | 0.243         | 0.151        | 0.191 |
| T2               | 0.215                                | 0.274         | 0.192        | 0.227 |
| Mean             | 0.198                                | 0.259         | 0.171        |       |
| LSD              | Period * Nano<br>0.022               | Nano<br>0.027 | Period<br>NS |       |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

### Plumule Length (cm)

The data in Table 3 indicate that treatment N1 had the greatest mean plumule length, at 11.625 cm, which was significantly higher than both N2 and N0 with a difference of 3.687 and 3.625 cm, respectively, though no significant difference was observed between the two treatments (Figure 5). Moderate nano-material concentrations are known to enhance nutrient availability and metabolic activity, leading to improved early seedling growth, as also reported by (2), which validated the beneficial consequences of nano-material seed treatments on this attribute. These results are in alignment with current outcomes in which the N1 treatment obtained an

optimal C–N–O ratio and higher plumule growth over lower and higher concentrations. These investigators demonstrated that nutrient levels positively affected this parameter via the enhancement of essential nutrients accumulated during plumule growth, mainly nitrogen (Figure 2).

Nitrogen is essential for the synthesis of necessary enzymatic proteins, which are involved in hydrolyzing stored carbohydrate and lipid compounds. It also helps to make up the nucleotides (DNA and RNA) which control rapid cell division in the plumule. Moreover, the N1 treatment showed specific equilibrium of carbon, oxygen, and nitrogen (Figure 3), which had a favorable effect on enhancement thereof (Figure 4). However, seed priming with higher dosages of nano amino acids led to a significant decrease in plumule length due to the toxicity effect of high level of nano-materials and nitrogen (Table 7) particularly when the elements are transformed into ammonium and nitrate. This reduction in plumule length could be due to the stress related effects probably in terms of ionic and osmotic balance as discussed in other studies. These results correspond to the observations of (9, 25), who noticed that the higher nano-materials did not have a positive effect on plant tissues.

For the soaking period, its effect was in accordance with the germination percentage and radicle length. T2 had a mean significant plumule length of 9.792 cm, a 1.209 cm difference from T1. This enhancement in T2 can be explained by the longer soaking duration, which allowed enough time for the seeds to attain the imbibition stage. At this stage dormant enzymes are activated and the formed cassette enzyme degrades stored carbohydrates and lipid. Furthermore, prolonged soaking periods also have a beneficial effect on nutrient availability and dry matter (Table 4), and seedlings with higher content of dry matter could obtain the larger plumule growth and stronger vigor (Figure 7). However, no significant difference was detected for this trait in the factor interaction process.

#### Seedling Dry Weight (g)

Treatment N1 of the nano amino acids recorded the highest mean at 0.259 g for this characteristic, significantly different from N2 and N0 by 0.088 g and 0.061 g, respectively (Table 4). The latter two treatments were also significantly different from each other. The higher seedling dry weight under N1 treatment could be due to the strong positive influence of this level on activating biochemical reactions inside the seed cells and its essential role in enhancing nutrient uptake efficacy, as well as the significant increase in water absorption since it recorded high values for nitrogen (Table 7). Also, the increased oxygen concentration (Table 9) due to better aeration as well as increased cellular activity contributed to increasing respiration and provided added ATP for cell division and tissue development.

This simultaneous physiological accommodation resulted in increased germination values (Table 1), enhanced photosynthetic efficiency and biomass accumulation, which in turn led to higher seedling dry weight (Table 4). These results agree with the observations by (7, 35, 42), who found that nano amino acids seed priming increased this trait. However, seedling dry weight decreased in response to increased levels of the acids, probably due to inhibition caused by their high concentrations on radicle length and plumule length which then negatively impacted this trait (Tables 2 and 3).

**Table 5.** Effect of nano amino acids seed priming on seedling vigor.

| Period<br>Treatment | Nano Treatment (mL L <sup>-1</sup> ) |               |                 | Mean   |
|---------------------|--------------------------------------|---------------|-----------------|--------|
|                     | N0(0)                                | N1 (0.75)     | N2 (1.50)       |        |
| T1                  | 9.408                                | 16.143        | 10.700          | 12.083 |
| T2                  | 14.270                               | 23.083        | 11.610          | 16.321 |
| Mean                | 11.839                               | 19.613        | 11.155          |        |
| LSD                 | Period * Nano<br>1.463               | Nano<br>1.792 | Period<br>2.535 |        |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

**Table 6.** Effect of nano amino acids seed priming on seedling vigor index.

| Period<br>Treatment | Nano Treatment (mL L <sup>-1</sup> ) |               |              | Mean  |
|---------------------|--------------------------------------|---------------|--------------|-------|
|                     | N0 (0)                               | N1 (0.75)     | N2 (1.50)    |       |
| T1                  | 12.11                                | 19.23         | 9.82         | 13.72 |
| T2                  | 16.98                                | 25.02         | 13.18        | 18.39 |
| Mean                | 14.54                                | 22.12         | 11.50        |       |
| LSD                 | Period * Nano<br>1.755               | Nano<br>2.149 | Period<br>NS |       |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

**Table 7.** Effect of nano amino acids seed priming on N atomic values in seedling tissue

| Period<br>Treatment | Nano Treatment (mL L <sup>-1</sup> ) |              |              | Mean  |
|---------------------|--------------------------------------|--------------|--------------|-------|
|                     | N0 (0)                               | N1 (0.75)    | N2 (1.50)    |       |
| T1                  | 3.55                                 | 10.45        | 14.02        | 9.34  |
| T2                  | 5.75                                 | 10.85        | 14.30        | 10.30 |
| Mean                | 4.65                                 | 10.65        | 14.16        |       |
| LSD                 | Period * Nano<br>NS                  | Nano<br>1.90 | Period<br>NS |       |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

**Table 8.** Effect of nano amino acids seed priming on C atomic values in seedling tissue.

| Period<br>Treatment | Nano Treatment (mL L <sup>-1</sup> ) |              |              | Mean  |
|---------------------|--------------------------------------|--------------|--------------|-------|
|                     | N0 (0)                               | N1 (0.75)    | N2 (1.50)    |       |
| T1                  | 51.50                                | 44.23        | 33.93        | 43.22 |
| T2                  | 56.53                                | 43.78        | 40.58        | 46.96 |
| Mean                | 54.01                                | 44.00        | 37.25        |       |
| LSD                 | Period * Nano<br>NS                  | Nano<br>4.86 | Period<br>NS |       |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

**Table 9.** Effect of nano amino acids seed priming on O atomic values in seedling tissue

| Period<br>Treatment | Nano Treatment (mL L <sup>-1</sup> ) |              |              | Mean  |
|---------------------|--------------------------------------|--------------|--------------|-------|
|                     | N0 (0)                               | N1 (0.75)    | N2 (1.50)    |       |
| T1                  | 40.20                                | 43.15        | 49.18        | 44.18 |
| T2                  | 40.83                                | 43.38        | 44.65        | 42.95 |
| Mean                | 40.51                                | 43.26        | 46.91        |       |
| LSD                 | Period * Nano<br>NS                  | Nano<br>5.12 | Period<br>NS |       |

Values are means of four replicates. LSD values were calculated at  $P \leq 0.05$ .

As for the soaking duration, the T2 treatment presented a higher mean dry weight (0.227 g), significantly different from T1 by 0.036 g. The better performance of T2 could be due to its higher radicle and plumule lengths (Tables 2 and 3), which favored the general of seedlings dry matter. However, no significant effect was detected for this characteristic in the interaction between factors.

#### Seedling Vigor

Table 5 shows that there were highly significant outcomes from the treatments and their interactions. The mean value of

N1 at 19.613 was the highest among all the treatments, and differed significantly by 8.458 and 7.774 from the N2 and N0 levels, respectively. However, seedling vigor was reduced at the upper concentration of N2. These results demonstrated that N1 was superior to N0 in this characteristic, which could be due to the higher germination percentage (Table 1) and its stronger positive effect on radicle and plumule lengths (Tables 2 and 3), all leading to superior seedling performance than the control group (Figure 7). These results are in accordance with those of (7, 34, 36).

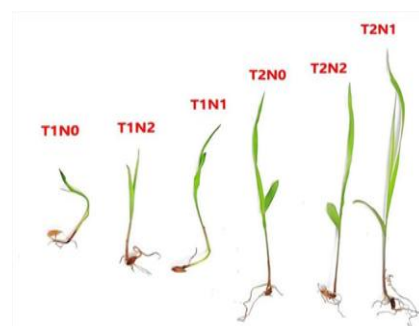
Results based on ANOVA (RCBD) showed that nano amino acids treatments, soaking duration, and their interactions significantly affected seedling vigor. Mean division using the LSD test ( $P \leq 0.05$ ) indicated that N1 produced the maximum seedling vigor compared to N2 and N0 (Table 5). The N1 nano amino acids treatment registered the highest mean value at 19.613, with significant differences of 8.458 and 7.774 over N2 and N0, respectively. However, seedling vigor decreased at high concentrations of N2.

The superiority of N1 in this characteristic might be interpreted in relation to its higher germination rate (Table 1) and positive effect on both root and shoot lengths (Tables 2, 3), which collectively contributed to better seedling performance (Figure 7). These findings are in line with (4, 9). Excesses can change the pH in the surrounding soil and remove the availability of key nutrients such as phosphorus, potassium or magnesium. This imbalance inhibits nutrient uptake and allows ions to accumulate environmentally toxic concentrations in plant tissues. Moreover, ROS produced through nanoparticles can lead to the destruction of cell membranes, proteins or nucleic acid (14).

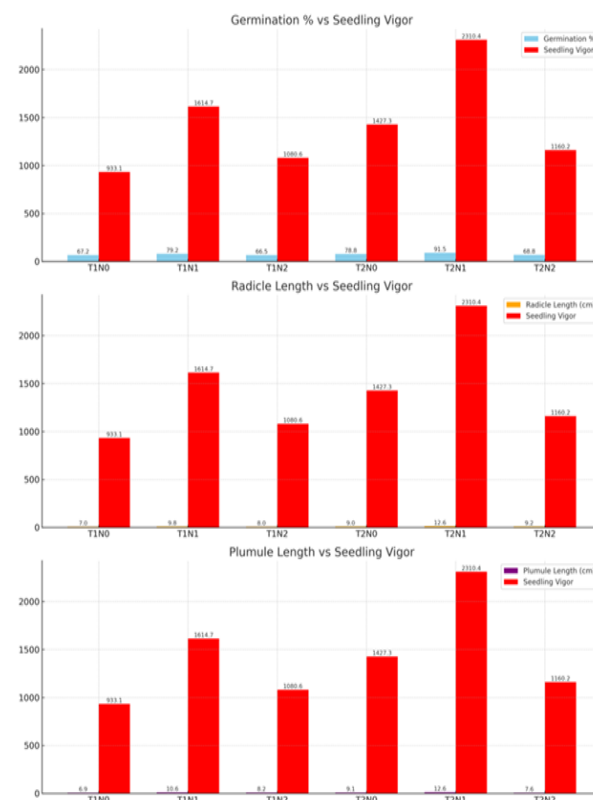
As a result, germination percent is reduced, radicle and plumule growth inhibited, and total seedling vigor diminished. The results are consistent with the observation of (9), who demonstrated that higher and lower concentrations of nanomaterials may have a positive and negative effect on germination, respectively. The findings show that increasing the concentration of nano amino acids decreased germination percentage and growth in radicle and plumule resulting in the total seedling vigor decreasing. Conversely, growth was stimulated at the lower concentrations, which corroborated with previous observations by (9). T2 demonstrated the highest average vigor index (16.321), significantly higher than T1 (LSD,  $P \leq 0.05$ ; Table 5), confirming the essential role of the duration of SO and nano amino acids-free solution concentrations at the period of seedling performance.

Concentrations of nano amino acids above 0.20% reduced the germination percentage, and radicle and plumule growth of seedlings, leading to lower total seedling vigor. Lower concentrations, by contrast, were stimulatory to growth. The vigor index, at 16.321, was highest with treatment T2, which significantly ( $P \leq 0.05$ ) exceeded the equivalent for T1 by 4.238 units (Table 5), according to ANOVA and followed by an LSD test. These results validate the important role of nano amino acids concentrations and priming duration in the optimization of seedling vigor.

The period of seed soaking or activation in the acids solution is significantly important to germination success and seedling vigor. If the soaking duration is too short, the seeds may not receive enough nutrients to facilitate faster and stronger germination. On the other hand, a soaking period that



**Figure 5.** Effect of nano amino acids seed priming treatments (T1N0–T2N2) on *Panicum maximum* L seedling growth after 10 days of germination.



**Figure 6.** Relationship between germination percentage, and radicle and plumule length with seedling vigor under seed priming and nano amino acids treatments.

is prolonged in a non-significant manner (10 h vs. 5 h) allows improved uptake of nano-elements such as nitrogen and oxygen (Tables 7 and 9). This induces enzymatic activity and increases metabolic efficiency, leading to greater radicle as well as plumule growth, and subsequently elevating seed vigor. Regarding the interaction, T2N1 presented the highest mean vigor index (23.083) and T1N0 the lowest (9.408).

### Seedling Vigor Index

As indicated in Table 6, the differences between nano amino acids treatments for seedling vigor index was significant, while there was no significant interaction recorded. The mean value for treatment N1 (22.12) was higher ( $p < 0.05$ ) compared to N2 and N0, though there was no significant difference for this parameter between N2 and N0 ( $P \leq 0.05$ ). The superiority of N1 is largely because of its higher values for germination percentage and seedling dry weight which are

the key components for seedling vigor. This dominance could be attributed to its higher germination percentage and seedling dry weight (Tables 1 and 4) which constitute the formula for calculating the seedling vigor index (Table 10).

This is further confirmed from Figure 12 which shows a clear relationship between germination percentage, total dry matter and vigor index. Moreover, N1 was the best among treatments in most of the characteristics assessed. The present results are similar to (7, 34, 36) which provided evidence of the beneficial effects of nano amino acids on seed vigor. On the contrary, this trait decreased at high levels of nano amino acids application as seen in N2. One possible explanation for this could reflect the adverse effect of superfluous nano amino acids on germination percentage (Table 1) and seedling dry weight (Figure 8), which are directly included in the computation of the seedling vigor index. As for soaking period, the mean vigor index of treatment T2 was significantly higher at 1.68 over that of T1 which recorded the lowest mean (13.72).

### Elemental Composition (SEM-EDS Analysis)

#### Nitrogen (N Atomic %)

The findings in Table 7 indicate that the N1 treatment (0.75 mL L<sup>-1</sup>) obtained the highest nitrogen content (10.65%) compared to the N0 (4.65%) and N2 treatments (14.15%). Moderate concentrations of nitrogen (N) in nutrient solutions tend to promote growth and improve photosynthetic efficiency, as N is an essential element for the synthesis of amino acids, proteins, and enzymes that contribute to plant growth [1]. At the highest concentration (N2), a relative decrease was noted, probably owing to nutrient saturation and mild toxic effects impeding optimal uptake (Figures 3 and 4). As can be seen from Table 1, the results are in good agreement with (4, 9, 37) which revealed the adverse effects of nano-materials in high doses. They found no significant differences for soaking period or interactive effects of these factors.

#### Carbon (C Atomic %)

The findings shown in Table 8 indicate that there were significant differences in carbon content between the nano amino acids treatments. Applying the N0 treatment caused this concentration to reach its maximum value (54.01%), whereas carbon content was significantly decreased under the N2 treatment (37.25%) as shown in figure (3). The reduced rate observed in this study could be attributed to the fact that excessive nano amino acids concentrations may favor plant uptake and consumption of formulated/accumulated carbon (mainly as carbohydrates) for energy and substantiate growth (39). Alternatively, excessive nano amino acids may destroy the balance of carbon metabolism. These results support the findings that over-dosing with nano amino acids decreased the efficiency of carbon fixation and increased plant dependence on respiration (18). No significant differences were observed for the soaking period or the interaction between factors.

#### Oxygen (O Atomic %)

For this characteristic, the data in Table 9 show that nano amino acids levels had a considerable contribution, and that soaking period as well as their interaction were non-significant. Mean oxygen content was significantly higher under treatment N2 (46.91%) than under treatment N0 (40.51%). An apparent increase in oxygen content may reflect

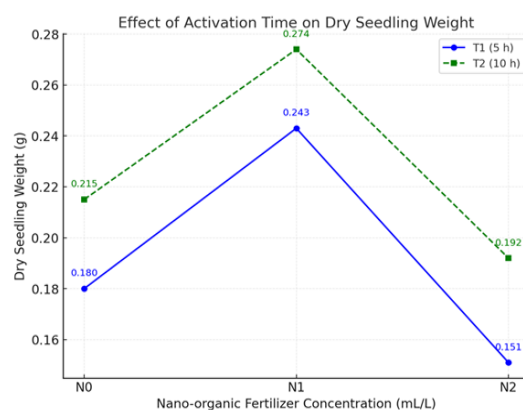


Figure 7. Effect of seed activation duration and nano amino acids concentration on seedling dry weight (g).

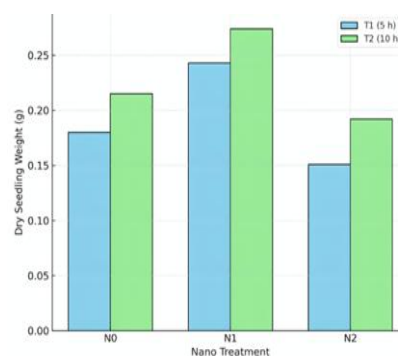


Figure 8. Effect of nano amino acids seed activation on dry seedling weight (g)

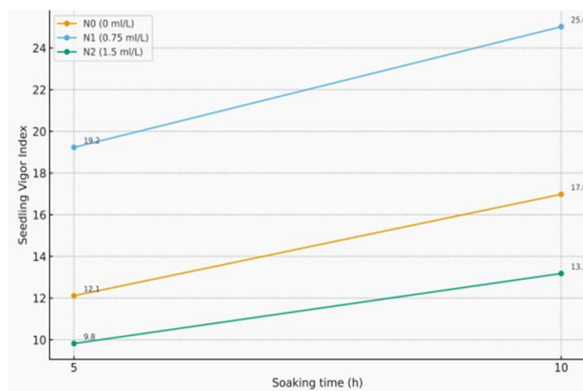


Figure 9. Line chart of seedling vigor index across soaking period

enhanced oxygen-associated metabolic activity; however, this is an indirect correlation, not a direct measurement. Nano-priming of seeds also enhances seed metabolism and signaling, triggering the development of nano-sized pores in seed-coats for better hydration. It also stimulates ROS as an antioxidant defense in the seed, resulting in increased biomass accumulation and entire seedling overall (11). A higher oxygen content may also be related to increased radicle and root-hair development, which in turn enhances the absorption of water and nutrients (Table 3).

**Table 10.** Comparison between seed activation treatments with nano amino acids (T1 × T2 × N0/N1/N2)

| Treatment | Soaking Conditions              | SEM-EDS Observations                                                   | Scientific Interpretation                           | Effect on Seedling Vigor           |
|-----------|---------------------------------|------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------|
| T1N0      | 5 h – distilled water (control) | Irregular distribution, large empty spaces, low N content              | Weak nutrient absorption → slow growth              | Lowest seedling vigor              |
| T1N1      | 5 h – 0.75 mL/L nano            | Clearer accumulation with uniform colors, better nutrient distribution | Moderate dose improved absorption and metabolism    | Noticeable increase in vigor       |
| T1N2      | 5 h – 1.5 mL/L nano             | Excessive accumulation → possible toxicity                             | High concentration led to ionic imbalance           | Reduced vigor compared to T1N1     |
| T2N0      | 10 h – distilled water          | Better nutrient absorption compared to T1N0                            | Longer soaking enhanced uptake without fertilizer   | Improved vigor compared to control |
| T2N1      | 10 h – 0.75 mL/L nano           | Very clear uniform accumulation of C, N, O                             | Balanced dose + longer soaking → optimal absorption | Highest vigor                      |
| T2N2      | 10 h – 1.5 mL/L nano            | Excessive accumulation, irregular distribution                         | High dose + long soaking → oxidative stress         | Reduced vigor                      |

## Conclusions

This investigation revealed the potential of nano amino acids (Optimus Plus) to significantly enhance germination performance and seedling vigor at the early growth stages of *Panicum maximum* L. The findings indicated that using the moderate concentration (N1 = 0.75 mL L<sup>-1</sup>) provided optimum results for all parameters, including germination percentage, radicle and plumule length, seedling dry weight, and vigor indices. SEM-EDS analysis proved that this application had even distribution of nitrogen, carbon, and oxygen in the seedling tissues which promotes nutrient uptake and metabolism activity respiratory levels.

Conversely, higher nano amino acids concentration (N2 = 1.5 mL L<sup>-1</sup>) induced osmotic and ionic stress which reduced germination and growth performance. Higher and longer priming periods (T2 = 10 hours) are also significant variables when compared to lower durations (T1 = 5 hours), as they enable full imbibition of water by seeds, activation of internal enzymes, and efficient uptake of nutrients. Both T2N1 (0.75 mL L<sup>-1</sup> nano amino acids with 10 h of priming) achieved the highest vigor index and the most sustainable physiological response. The reported effects are restricted to the seed germination and early seedling stage of *Panicum maximum* L. and not generalized to other species or growth stages.

## Supplementary Materials

No supplementary materials.

## Author Contributions

Ahmed Chyad Al-Fahad: methodology, writing—original draft preparation; Alaa Abdulghani Hussein: review and editing. Both authors have read and agreed to the published version of the manuscript.

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## Informed Consent Statement

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No data availability statement.

## Conflicts of Interest

The authors declare no conflict of interest in the conducting of this research.

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